PIZZA SALES ANALYSIS

# Business Intelligence Using SQL

A restaurant analytics case study translating raw order data into revenue, demand, and product strategy insights.

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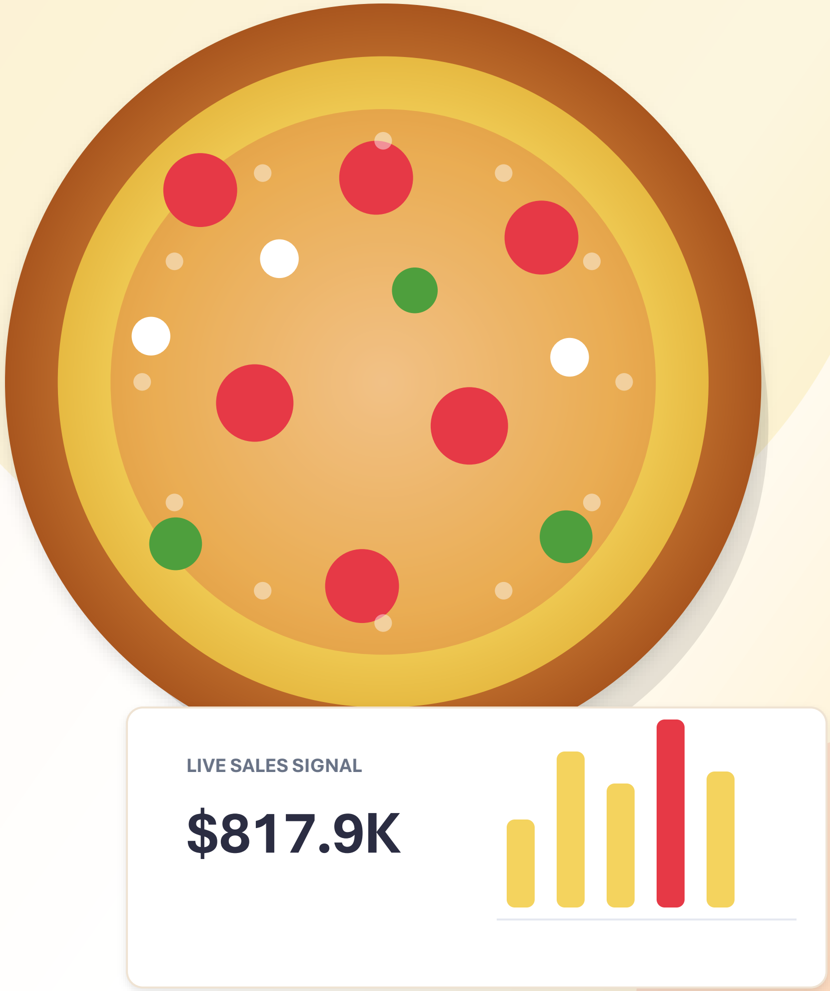



SQL -> Analysis -> Business Decisions

EXECUTIVE SUMMARY

# A high-volume pizza operation with balanced category revenue

Annual sales data from 2015 shows strong product-market fit, clear daypart peaks, and a menu mix with no single category over-concentrated.

| TOTAL ORDERS | TOTAL REVENUE | AVG ORDER VALUE | TOP PIZZA | TOP CATEGORY |
|---|---|---|---|---|
| **21,350** | **$817.9K** | **$38.31** | **Thai Chicken** | **Classic** |
| Transactions captured in the orders table | Generated across 48,620 line items | Revenue per order | $43.4K revenue leader | 26.9% of revenue |

### Management takeaway

Revenue is broad-based across four categories, while demand concentrates into two daily rush windows. The best opportunity is operational: protect best sellers, staff around peaks, and use category-specific bundles to raise basket size.

**49,574 pizzas sold**

BUSINESS PROBLEM

# Management needed a revenue and demand playbook

The dataset answers the questions restaurant managers ask before adjusting menus, staffing, promotions, and inventory.

$ **GROW REVENUE**

### Which categories and pizzas generate the most sales value?

Prioritize menu placement, bundles, and premium offers.

MENU **OPTIMIZE MENU**

### Which products sell frequently and which contribute less?

Protect hero products and tune category-level promotions.

TIME **PLAN DEMAND**

### When do customers order most during the day?

Align staffing, prep, and inventory with lunch and dinner peaks.

**Analytical framing: turn transactional SQL outputs into decisions on product mix, daily operations, and promotional strategy.**

DATASET OVERVIEW

# Four connected tables power the sales analysis

The dataset combines order headers, line items, pizza variants, and product attributes for a complete restaurant transaction model.

**orders**

**21,350 rows**

order_id | date | time

**order_details**

**48,620 rows**

order_details_id | order_id | pizza_id | quantity

**pizzas**

**96 rows**

pizza_id | pizza_type_id | size | price

**pizza_types**

**32 rows**

pizza_type_id | name | category | ingredients

**Orders**
transaction header

→

**Order Details**
line item quantity

→

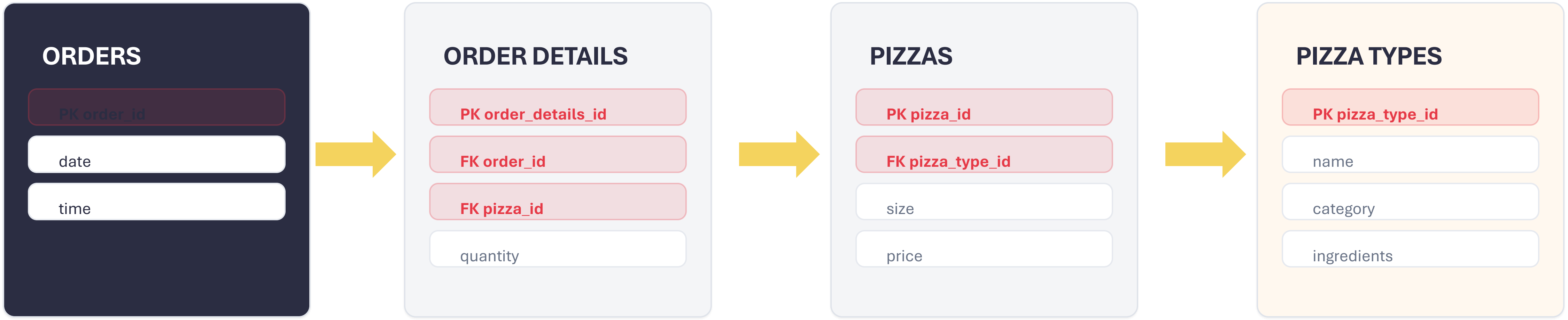
**Pizzas**
size + price

→

**Pizza Types**
name + category

# The ERD connects orders to product attributes

Every business metric is created by joining order timing, quantity, price, and pizza category dimensions.




**Join logic: orders.order_id -> order_details.order_id -> pizzas.pizza_id -> pizza_types.pizza_type_id**

SQL SKILLS DEMONSTRATED

# The analysis moves from joins to advanced ranking

The SQL work solves business questions by combining relational modeling, aggregation, and windowed analytics.

JOIN

### JOINs

Connected 4 tables for full transaction context.

SUM

### Aggregations

SUM, COUNT, AVG, ROUND for KPIs and sales totals.

GB

### GROUP BY

Category, pizza, date, and hour-level summarization.

SUB

### Subqueries

Total revenue denominator and daily average logic.

CTE

### CTEs

Reusable layer for category-level ranking output.

WIN

### Window Functions

Cumulative revenue and ranked products.

RANK

### Ranking

Top 3 pizzas by category using RANK().

BI

### Business Translation

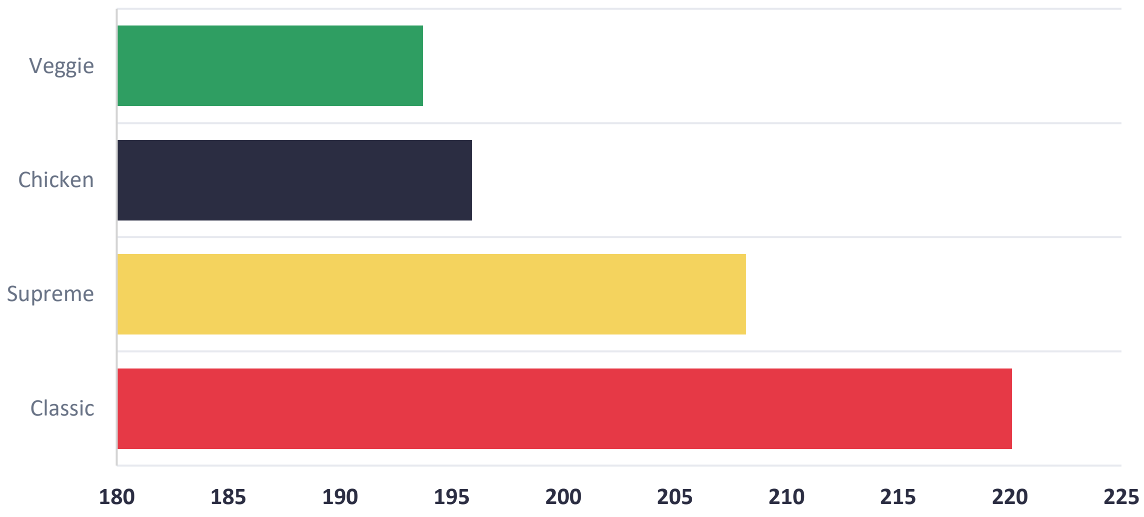
SQL outputs converted into strategy insights.

REVENUE ANALYSIS

# Classic leads, but the category portfolio is balanced

All four categories contribute meaningfully, reducing dependency on a single menu segment.

## Revenue by category




## Revenue contribution

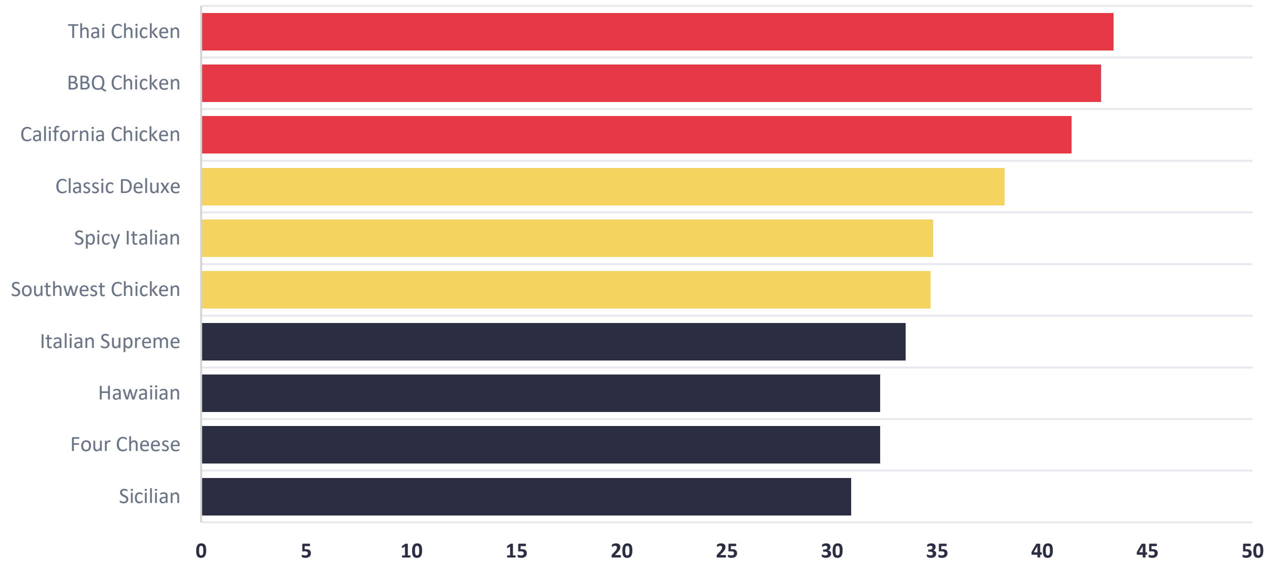
- **Classic: 26.9%**
- Supreme: 25.5%
- Chicken: 24.0%
- Veggie: 23.7%

Insight: Classic is the strongest revenue driver, but the narrow spread across categories suggests a healthy, diversified menu.

# Chicken pizzas dominate the top revenue rankings

The top three revenue products are all Chicken category pizzas, while Classic Deluxe is the highest-volume pizza.

## Top 10 pizzas by revenue




**#1** **Thai Chicken Pizza**
$43.4K | 2,371 units | Chicken

**#2** **Barbecue Chicken Pizza**
$42.8K | 2,432 units | Chicken

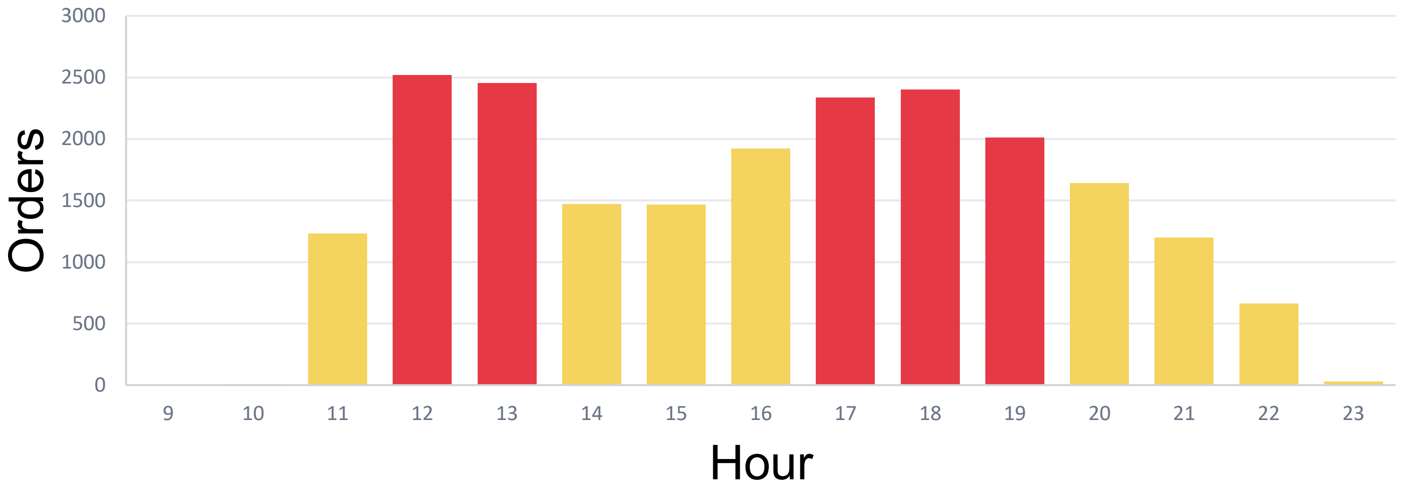
**#3** **California Chicken Pizza**
$41.4K | 2,370 units | Chicken

**Volume leader: Classic Deluxe, 2,453 units**

# Orders spike at lunch and again during dinner

Demand is concentrated around 12:00-13:00 and 17:00-19:00, creating clear prep and staffing windows.

## Orders by hour of day




### Lunch rush

**12:00-13:00 | 4,975 orders**

Prep high-turnover toppings before noon.

### Dinner rush

**17:00-19:00 | 6,744 orders**

Schedule kitchen coverage through early evening.

### Low demand

**09:00-10:00 / 23:00 | 37 orders**

Use low-volume windows for prep and cleanup.

CATEGORY PERFORMANCE

# Classic wins on both volume and value

Category-level comparison shows a strong Classic base and a balanced revenue spread across Supreme, Chicken, and Veggie.

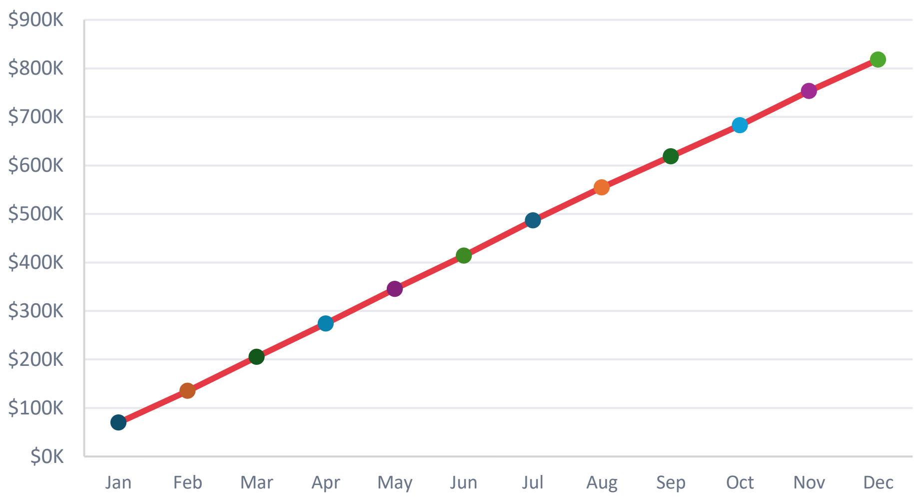
| | | | |
|---|---|---|---|
| **Classic**<br>8 pizza types | **$220.1K**<br>26.9% revenue | Revenue index | 14,888 units |
| **Supreme**<br>9 pizza types | **$208.2K**<br>25.5% revenue | Revenue index | 11,987 units |
| **Chicken**<br>6 pizza types | **$195.9K**<br>24.0% revenue | Revenue index | 11,050 units |
| **Veggie**<br>9 pizza types | **$193.7K**<br>23.7% revenue | Revenue index | 11,649 units |

**Business implication: keep Classic products visible, then use targeted promotions to lift lower-share categories without weakening menu balance.**

# Window functions turn transactions into trend and ranking insight

CTEs and window functions were used to analyze cumulative revenue and identify top products within each category.

## Cumulative revenue over time




**SQL pattern: SUM(revenue) OVER (ORDER BY order_date)**

## Top 3 products by category

### Chicken

| | |
|---|---|
| **1. Thai Chicken** | **$43.4K** |
| 2. BBQ Chicken | **$42.8K** |
| 3. California Chicken | **$41.4K** |

### Classic

| | |
|---|---|
| **1. Classic Deluxe** | **$38.2K** |
| 2. Hawaiian | **$32.3K** |
| 3. Pepperoni | **$30.2K** |

### Supreme

| | |
|---|---|
| **1. Spicy Italian** | **$34.8K** |
| 2. Italian Supreme | **$33.5K** |
| 3. Sicilian | **$30.9K** |

### Veggie

| | |
|---|---|
| **1. Four Cheese** | **$32.3K** |
| 2. Mexicana | **$26.8K** |
| 3. Five Cheese | **$26.1K** |

**SQL pattern: CTE + RANK() OVER (PARTITION BY category ORDER BY revenue DESC)**

KEY FINDINGS

# Five insights turn the SQL output into business action

The project surfaces clear revenue drivers, product leaders, category strength, and operational demand patterns.

REVENUE BASE

## $817.9K

Annual revenue from 21,350 orders.

TOP REVENUE DRIVER

## Thai Chicken

$43.4K revenue and 2,371 units.

MOST POPULAR PIZZA

## Classic Deluxe

2,453 units, highest volume product.

STRONGEST CATEGORY

## Classic

26.9% revenue share and 14,888 units.

PEAK ORDERING HOURS

## 12-13 & 17-19

Two rush windows shape staffing needs.

# Use the findings to improve menu, staffing, and basket size

The strongest next steps are practical restaurant-management actions tied directly to observed sales patterns.

## Feature hero products

Keep Thai Chicken, BBQ Chicken, and California Chicken prominent in menus and campaigns.

**Expected impact: protect high-value demand.**

## Build lunch bundles

Pair Classic products with drinks or sides during the 12:00-13:00 demand peak.

**Expected impact: lift AOV in the busiest window.**

## Staff around two rushes

Schedule prep and kitchen coverage for 11:30-13:30 and 16:30-19:30.

**Expected impact: faster service and fewer stockouts.**

## Tune category promotions

Use Veggie and Supreme promotions during softer hours without discounting best sellers unnecessarily.

**Expected impact: smoother demand and better mix.**

**Decision principle: prioritize actions that convert SQL insights into observable restaurant operations.**

WHAT I LEARNED

# SQL becomes valuable when it supports business decisions

This project strengthened both technical querying and analytical storytelling skills.

1

### Model

Understood relational tables and keys.

2

### Join

Built complete transaction context.

3

### Measure

Created revenue, quantity, and AOV KPIs.

4

### Rank

Used CTEs and windows for top products.

5

### Tell

Converted outputs into BI recommendations.

Skills developed: relational data modeling, SQL aggregation, contribution analysis, cumulative trend analysis, product ranking, dashboard thinking, and business storytelling.

THANK YOU

# Pizza Sales Analysis

Business Intelligence Using SQL

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Portfolio-ready SQL case study | 2015 restaurant sales dataset